Local Service and Business Intelligence Database

Local Service Insights • Columbia, Missouri automotive services
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Purpose, scope, and evidence

This database helps students and residents find a recorded automotive service provider in Columbia and trace the offering to its source. We include a verifiable business name, one physical Columbia street location, and at least one explicit service. Extra branches and repeated listings do not add businesses. The convenience sample deliberately records a bounded set of offerings rather than every service each provider supplies.

Businesses	Services	Categories	Offerings	Source pages
15	4	2	20	20

The 20 source pages have 18 publisher labels. Provider sites supply the service evidence. Three address checks compare All-Star with AAA (S04/S18), Midas with Michelin (S09/S19), and Walt's with BBB (S13/S20). These organizations publish separate listings; the checks do not claim independent on-site measurement or unrelated ownership for every provider brand. Exact URLs, dates, values, and raw IDs appear in the source and corroboration registers.

The unchanged manual capture contains 45 observations: 15 identity/location rows, 20 offerings, three corroborations, and seven review/exclusion rows. AI-assisted collection accessed the original pages on October 4, 2026. Cleaning standardizes identity/location wording and maps explicit service wording to controlled types. Five real before/after examples are saved. A repeated All-Star footer becomes one business, and provider-confirmed old names merge into Dobbs and Norris. Two extra branches are excluded. Tom's conflicting alignment claims are excluded rather than guessed. See data/cleaning_note.md and the unchanged data/raw/observations.csv.

Four business rules

1. Identity: Each distinct business has one stable ID and one selected physical location. Duplicate listings and additional branches do not add businesses.

2. Relationships: Each service belongs to exactly one category. Businesses and services have a many-to-many relationship. Every included business has at least one recorded offering.

3. Required values: Names, selected address/city, service/category names, and source/raw references are required. An unverified website is SQL NULL.

4. Duplicate prevention and evidence: Store each business/service pair once. Link business identity, location, and offering facts to supporting sources and saved observations.

Model and implementation

The five tables are Business, ServiceCategory, Service, BusinessService, and Source. BusinessService resolves the many-to-many relationship with composite key (business_id, service_id). Service.category_id models one-to-many category membership. Source foreign keys preserve provenance. Separate identity and location source references allow two supporting pages when necessary. The selected address stays in Business because a branch subsystem is outside scope.

Primary and foreign keys, NOT NULL, a unique selected-location tuple, and RESTRICT deletion/update behavior enforce the represented rules. T1 verifies the final-sample requirement of an offering for every business. Website is the only nullable field. README lists each table's purpose/keys; 00_schema.sql documents column types. The complete ER diagram is page 3. General brand evidence means business offering; branch availability unverified.

Design reasoning and observed results

Normalization to 3NF

Begin with a deduplicated combined relation R: business ID/name/address/city/website and identity/location evidence IDs, service ID/name, category ID/name, offering evidence IDs, and source metadata for each evidence role. One candidate key is (business_id, service_id), assuming one selected supporting source per offering. The unchanged raw capture may contain multiple observations; it is not this deduplicated relation.

Determinant	Dependent facts
business_id	Name, selected address/city, website, business evidence
service_id	Service name, category_id
category_id	Category name
business_id, service_id	Offering source_id and raw_row_id
source_id (each role)	Publisher, title, URL, collection date, raw-file locator

Business and service details depend on only part of the offering key. Category and source metadata create further transitive dependencies. Decompose R into Business, Service, ServiceCategory, BusinessService, and Source. Under these dependencies, every nontrivial determinant in each final relation is a key, so the relations satisfy 3NF. The same Source table serves identity, location, and offering roles through separate foreign keys.

This removes an update anomaly: a repeated business address could change on one offering row but remain stale on another. Business now stores it once. Joins reconnect BusinessService to Business and Service, Service to ServiceCategory, and each evidence reference to Source. Alias Source separately when reconstructing its three roles. Foreign keys enforce those links.

Two findings supported by saved query outputs

Finding 1 (Q2): Columbia and service ID 1 (Oil change) return 13 distinct recorded providers. For example, Jiffy Lube is business 10 with offering O10 and source 10. The full provider list, source URLs, and raw references are in evidence/Q2.csv. The independent CSV check confirms the complete provider-ID set and no duplicate provider rows.

Finding 2 (Q3): At a minimum of two providers, Oil change has 13 and Brake service has 5. The four manual provider counts are 13, 5, 1, and 1 for oil, brakes, tire rotation, and alignment respectively. The one-provider groups do not pass HAVING >= 2. See evidence/Q3.csv and query_checks.md.

Validation, limitations, and follow-up

MySQL 8.0.46-0ubuntu0.24.04.3 executed both clean builds with foreign keys active. Ordered table exports and all six outputs match byte for byte. T1 passes counts/evidence checks. T2 rejects a nonexistent-business reference (1452), and T3 rejects a repeated pair (1062). T4 separately proves NOT EXISTS, LEFT JOIN, missing-website, and manual-count behavior, then removes the fixture. Socket restrictions required offline MySQL initialization execution and identical SELECT exports; the submitted normal scripts rebuild without that harness. Actual logs are in evidence/test_log.md.

Collection favors accessible websites and is capped at 20 offerings. Missing offerings mean not recorded, not absent. All 15 real businesses have verified websites, so Q6 is honestly empty; T4 tests NULL separately. Provider counts do not establish unmet demand. A useful follow-up is: How would the recorded distribution change if we collected a complete service inventory for every sampled provider?

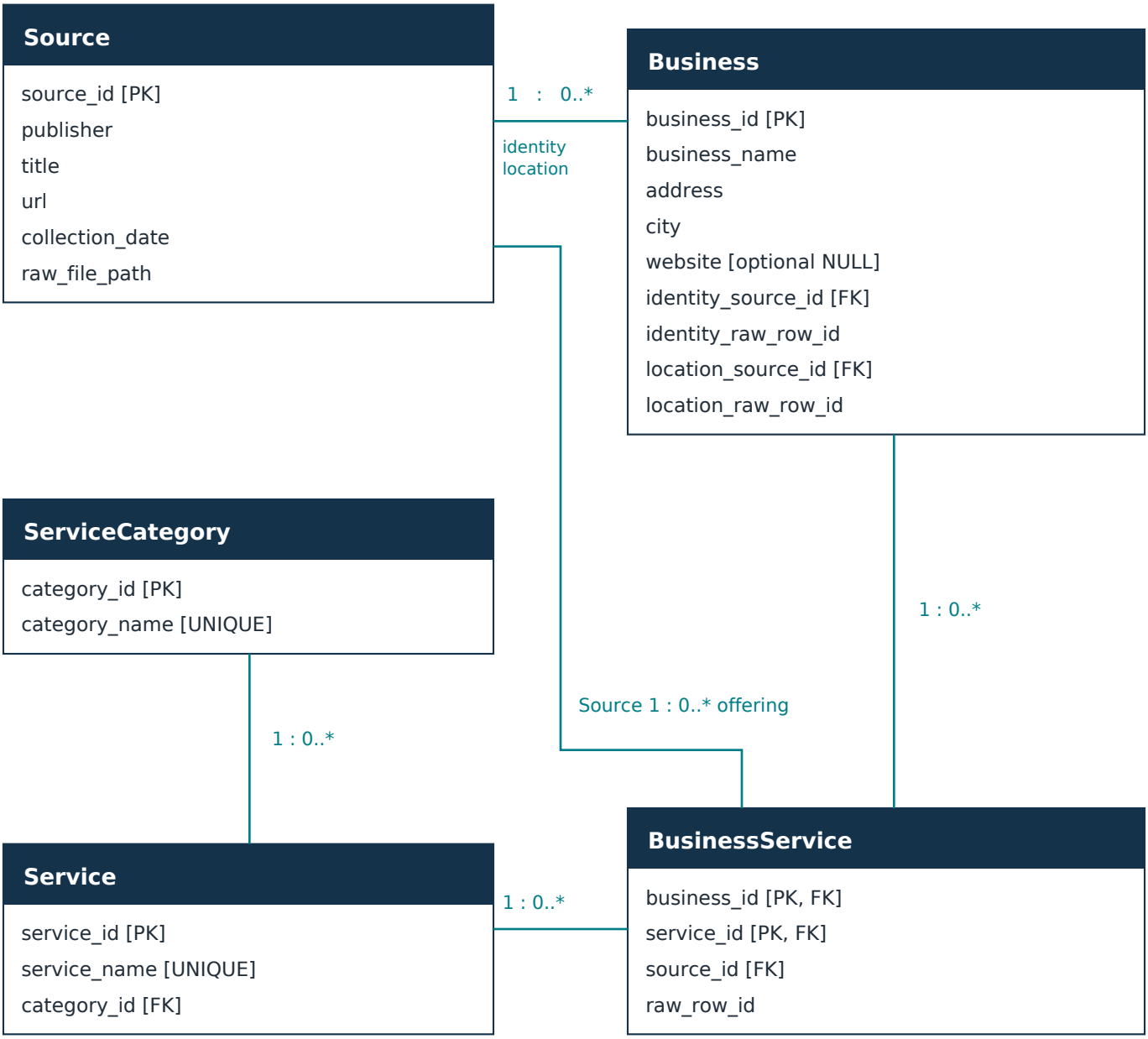
Responsibility and AI use

Mayowa supplied the request and roster. ChatGPT assisted source collection/comparison, classification, SQL, actual MySQL validation, and document preparation. Completed student reviews or recordings are not claimed. README documents the AI assistance and planned member responsibilities. Every member must review the evidence and demonstrate personal understanding in the video. The four-person roster requires instructor approval because the brief specifies 2–3.

Evidence: source IDs refer to data/clean/sources.csv; raw IDs refer to data/raw/observations.csv. Data/corroboration.csv, data/cleaning_examples.csv, README, sql/, and evidence/ provide the supporting details. No live downloads are needed.

ER diagram and participation

Each child FK requires exactly one existing parent. Parents can exist before child rows load, so structural relationships allow 0..* children. The final sample requires at least one offering per Business, validated by T1. All attributes are required except website. Identity and location are two mandatory Source relationships.



PK / FK participation and composite constraints

BusinessService PK = (business_id, service_id). Business UNIQUE = (name, address, city).